THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** scalar	Magnitude only.
2	**(d)** push or pull	Force.
3	**(c)** force x displacement	W = Fs.
4	**(c)** vibration	Vibration.
5	**(c)** displacement	Area.
6	**(a)** more inertia	Mass.
7	**(c)** changes the direction of force	MA = 1.
8	**(a)** above 20 kHz	Higher.
9	**(c)** 50 km/h	100/2.
10	**(c)** 2 m/s squared	F/m.
11	**(a)** 20 J	Fs.
12	**(d)** 100 oscillations per second	Definition.
13	**(d)** vector	Magnitude and direction.
14	**(c)** newton	N.
15	**(b)** joule	J.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Total length of the path travelled (2).	2
17	A body remains at rest or in uniform motion unless a force acts (2).	2
18	Force multiplied by displacement in the direction of the force (2).	2
19	A form of energy produced by vibrating bodies (2).	2
20	Rate of change of velocity (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Scalar and vector (1); Distance is always positive (1); Displacement may be zero (1).	1+1+1
22	Dust from carpet (1); Passengers falling backward (1); Coin on a card (1).	1+1+1
23	Moving car (1); Flowing water (1); Flying bird (1).	1+1+1
24	Tuning fork (1); Drum (1); Vocal cords (1).	1+1+1
	OR: 108 x 5/18 (1); = 30 (1); m/s (1).	1+1+1
25	F = ma (1); 1000 x 3 (1); 3000 N (1).	1+1+1
	OR: W = Fs (1); 20 x 5 (1); 100 J (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Distance (1); Displacement (1); Speed (1); Velocity (1); Acceleration (1).	1+1+1+1+1
	OR: First law (1); Second law (1); Third law (1); Examples (1); Formulas (1).	1+1+1+1+1
27	Force (1); Displacement (1); In the direction of force (1); Examples (1); Zero work cases (1).	1+1+1+1+1
	OR: Vibration (1); Medium (1); Compression (1); Rarefaction (1); Wave travels (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** medium	To travel.
2	**(c)** acceleration	Slope.
3	**(b)** third law	Action-reaction.
4	**(b)** pivot of a lever	Fixed point.
5	**(b)** below 20 Hz	Lower.
6	**(a)** 2 m/s	10/5.
7	**(a)** 6 N	ma.
8	**(b)** 20 W	W/t.
9	**(d)** 0.1 s	1/f.
10	**(b)** distance/time	Formula.
11	**(d)** inertia	Law of inertia.
12	**(a)** watt	W.
13	**(b)** vacuum	No medium.
14	**(d)** zero	Back at start.
15	**(b)** inertia	Inertia of rest.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Product of mass and velocity (2).	2
17	Energy is neither created nor destroyed (2).	2
18	Distance between two consecutive compressions or crests (2).	2
19	s = ut + (1/2)at squared (1); Meaning of symbols (1).	1+1
20	Mass in kg, weight in N (1); Weight varies with g (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	T = 1/f (1); 1/250 (1); 0.004 s (1).	1+1+1
22	Scalars: distance, speed, time (1); Vectors: displacement, velocity, acceleration (1); Direction is the difference (1).	1+1+1
23	Subtract (1); 10 - 4 (1); 6 N east (1).	1+1+1
24	PE to KE (1); Total constant (1); Without air resistance (1).	1+1+1
	OR: Increases with temperature (1); Warm air (1); Cold air slower (1).	1+1+1
25	Scalar vs vector (1); Distance vs displacement (1); Speed is never negative (1).	1+1+1
	OR: Passenger moving forward when brakes are applied (1); Athlete running after finishing (1); Fruits falling when a tree is shaken (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Total distance 100 km (1); Total time 3 h (1); 33.3 km/h (1); Formula (1); Units (1).	1+1+1+1+1
	OR: Fuel burns (1); Gases expelled downward (1); Reaction pushes up (1); Third law (1); Space travel (1).	1+1+1+1+1
27	Ramp (1); Longer path (1); Less force (1); MA = l/h (1); Examples (1).	1+1+1+1+1
	OR: Audible (1); Infrasonic (1); Ultrasonic (1); Examples (1); Uses (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** first class lever	Fulcrum in the middle.
2	**(a)** reflection of sound	Reflection.
3	**(b)** 5 m/s squared	10/2.
4	**(c)** 100 N	mg.
5	**(c)** 100 J	mgh.
6	**(a)** wavelength	v = f lambda.
7	**(c)** displacement/time	Formula.
8	**(a)** mass	Mass.
9	**(b)** work/time	Rate.
10	**(a)** longitudinal wave	Compressions.
11	**(d)** 5/18 m/s	1000/3600.
12	**(d)** backward	Inertia of rest.
13	**(b)** second class lever	Load in the middle.
14	**(b)** 0.1 s	Persistence.
15	**(b)** 100 m	20 x 5.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Rigid bar that turns about a fixed point (2).	2
17	SONAR (1); Medical imaging (1).	1+1
18	Shortest distance between initial and final positions with direction (2).	2
19	Force equals mass times acceleration (rate of change of momentum) (2).	2
20	Rate of doing work (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Stretched bow (1); Water in a tank (1); Book on a shelf (1).	1+1+1
22	Direction of vibration (1); Compressions vs crests (1); Sound vs rope waves (1).	1+1+1
23	20 x 18/5 (1); = 72 (1); km/h (1).	1+1+1
24	p = mv (1); 5 x 4 (1); 20 kg m/s (1).	1+1+1
	OR: First: scissors (1); Second: wheelbarrow (1); Third: tweezers (1).	1+1+1
25	SONAR (1); Bats finding prey (1); Depth of sea (1).	1+1+1
	OR: Pythagoras (1); sqrt(16 + 9) (1); 5 m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	a = (v - u)/t (1); at = v - u (1); v = u + at (1); Symbols (1); Conditions (1).	1+1+1+1+1
	OR: Definition (1); Rest (1); Motion (1); Example 1 (1); Example 2 (1).	1+1+1+1+1
27	W = Fs (1); F = ma (1); v squared = 2as (1); W = (1/2) m v squared (1); Equal to KE (1).	1+1+1+1+1
	OR: Crest (1); Trough (1); Amplitude (1); Wavelength (1); Frequency (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.